

1 • Break the problem down (in this order)

- Name it:** write " $X = \dots \sim \dots$ ". "Completely random / at rate" $\Rightarrow$ Poisson process; "no spot favoured", random point/time/cut $\Rightarrow$ uniform; sums/heights/returns $\Rightarrow$ normal; "until first" $\Rightarrow$ geometric; "until r th" $\Rightarrow$ NegBin.
- Count / $\mathbb{E}$ of a count** $\Rightarrow$ indicators, $\mathbb{E} = \sum P$ (dependence irrelevant). **Equal chances** $\Rightarrow$ symmetry / bijection (swap, reverse, complement, $x \mapsto c - x$); "which of k is first/largest" $= \frac{1}{k}$.
- "Until" / repeated rounds** $\Rightarrow$ states + first-step: $r(s) = \sum_{s'} P_{ss'}[c + r(s')]$ (prob $c = 0$, time $c = 1$, best = max, game = not). Same rules every turn $\Rightarrow$ drop the turn index; deadline $\Rightarrow$ keep it. State = only what the stop-check needs (run length, last roll, distance, parity).
- DP forms:** $i \rightarrow$ end $\cdot$ ending at $i \cdot$ pair $(i, j) \cdot$ interval $(l, r) \cdot$ game win $(s) = \exists$ move to losing.
- Self-loop** $x = c + qx \Rightarrow x = \frac{c}{1-q}$. **A before B** (per round a, b , rest "nothing"): $P = \frac{a}{a+b}$, rounds $\sim \text{Geo}(a + b)$. **Alternating** (hit p_1 , then p_2): starter wins $\frac{p_1}{1-q_1q_2}$; equal $p: \frac{1}{2-p}$.
- Conditioned "until"** (given no 5 before the 6, etc.): $\mathbb{E}[X|A] = \frac{\mathbb{E}[X1_A]}{P(A)}$; recurse both. "Given A came before B": time to first of A u B is $\text{Geo}(a + b)$ independent of which won.
- Max/min** $\Rightarrow$ CDF: $P(\max \leq x) = \prod F_i$, $P(\min > x) = \prod (1 - F_i)$. **$\mathbb{E}$ of nonneg integer** $\Rightarrow \sum_{k \geq 1} P(N \geq k)$.
- Stuck?** $n = 1, 2, 3$, guess, prove; complement; extreme case; invariant.
- Wording $\rightarrow$ tool:** "expected number of" $\rightarrow$ indicators §4 $\cdot$ "before" $\rightarrow$ A-before-B / Penney §9 $\cdot$ "in a row / pattern" $\rightarrow$ Conway §9 $\cdot$ "reaches $\pm$ / ruin" $\Rightarrow$ §10 $\cdot$ "returns / long run" $\rightarrow$ Kac §11 $\cdot$ "random point, cut, time" $\rightarrow$ §6 $\cdot$ "at rate, arrive" $\rightarrow$ §7 $\cdot$ "may stop / optimal" $\rightarrow$ backward §13 $\cdot$ "who wins" $\rightarrow$ P/N §14 $\cdot$ "can it ever" $\rightarrow$ invariant §15 $\cdot$ "worst case, minimum tries" $\rightarrow$ information bound §15 $\cdot$ "given that" $\rightarrow$ shrink the space §3.

2 • Counting

- ordered/rep $n^r \cdot$ ordered $\frac{n!}{(n-r)!} \cdot$ subsets $\binom{n}{r} \cdot$ words $\prod_{k=1}^n n^k$ (order of taking from stacks = word of stack labels) $\cdot$ circle $(n-1)!$, necklace $\frac{(n-1)!}{2}$.
- Stars & bars** $x_1 + \dots + x_k = n: \geq 0 \Rightarrow \binom{n+k-1}{k-1}; \geq 1 \Rightarrow \binom{n-1}{k}; \geq a_i: n \rightarrow n - \sum a_i; \leq n$: add slack. Non-decreasing length r from m values $\binom{m+r-1}{r}$; strictly increasing $\binom{m}{r}$.
- Runs:** R items into k nonempty blocks $\binom{R-1}{k-1}$. Lattice path a, R, b, U with exactly t turns: $t + 1$ alternating blocks, start R or U.
- Gaps:** k of n in a row, none adjacent $\binom{n-k+1}{k}$; circle $\frac{n-k}{n} \binom{n-k}{k}$. Together $\Rightarrow$ glue (block $\times$ internal order); apart $\Rightarrow$ complement or insert into gaps.
- PIE** $|\bigcup A_i| = \sum (-1)^{k+1} \sum_{|S|=k} |A_S|$; symmetric: $\sum_k (-1)^{k+1} \binom{n}{k} |A_{1..k}|$. Pairs adjacent (m couples, $2m$ seats row): $|A_{1..k}| = 2^k (2m - k)!$. Onto $n \rightarrow k: \sum_j (-1)^j \binom{k}{j} (k - j)^n$. Exactly 2 of 3: $\sum |A_{ij}| - 3|A_{123}|; \geq 2: -2|A_{123}|$. Divisible by some of $\{a, b, c\}$: intersections = lcm.
- Derangements** $D_n = n! \sum_k \frac{(-1)^k}{k!} \approx \frac{n!}{e}$, $D_n = (n-1)(D_{n-1} + D_{n-2})$: 1, 0, 1, 2, 9, 44, 265, 1854. Exactly k fixed: $\binom{n}{k} D_{n-k}$.
- Identities:** Vandermonde $\sum \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r} \Rightarrow \sum \binom{n}{2} = \binom{2n}{n}$, $\sum \binom{a}{b} \binom{b}{c} = \binom{a+b}{b-1}$; hockey $\sum_{j \leq n} \binom{j}{r} = \binom{n+1}{r+1}$; $\sum k \binom{n}{k+1} = n2^{n-1}$; $\sum_{\text{odd}} = \sum_{\text{even}} = 2^{n-1}$; $\sum_{k=1}^n \binom{n}{k} = \frac{1}{n+1} \binom{n+1}{n+1}$. Largest antichain of subsets $\binom{n}{\lfloor n/2 \rfloor}$.
- Conjugate trick:** $(a + \sqrt{b})^n + (a - \sqrt{b})^n \in \mathbb{Z}$; if $|a - \sqrt{b}| < 1$: $(a + \sqrt{b})^n$ is just below an integer (decimals .999... , n even).
- Paths/walks:** $(0, 0) \rightarrow (a, b) \binom{a+b}{a}; \pm 1$ walk $0 \rightarrow d$ in $n: \binom{n+d}{(n+d)/2}$. **Reflection:** paths touching $m \leftrightarrow$ paths ending at $2m$ —end. Never below 0 ($u \geq d$): $\binom{u+d}{d} - \binom{u+d}{d-1}$. **Ballot** A always ahead $\frac{a-b}{a+b}$. Catalan C_n : 1, 1, 2, 5, 14, 42, 132, 429 (brackets, Dyck, triangulations, binary trees). Two

walkers from opposite corners of $n \times n$ (up/right vs down/left): meet w.p. $\binom{2n}{n} / 4^n$.

- Recurrences:** jumps of 1 or 2 to n : F_{n+1} ; jumps ≥ 2 : F_{n-1} ; odd jumps: F_n (trick: subtract the $n - 1$ line). Bits, no "11": F_{n+2} ; on a ring of n : $L_n = F_{n-1} + F_{n+1}$ (1, 3, 4, 7, 11, 18, 29, 47). m -ary no "00": $b_n = (m - 1)(b_{n-1} + b_{n-2})$. Roots $r \neq s$: $A r^n + B s^n$; double $(A + B n) r^n$.
- Geometry counts:** n -gon diagonals $\frac{n(n-3)}{2}$; interior crossings $\binom{n}{4}$; regions: n lines $1 + n + \binom{n}{2}$, n circles $n^2 - n + 2$, n planes $\binom{n}{0} + \dots + \binom{n}{3}$, chords of n circle points $1 + \binom{n}{2} + \binom{n}{4}$. Regular n -gon two random segments meet (share vertex or cross): $\frac{n \binom{n-1}{2} + \binom{n}{2}}{\binom{n}{2}}$.
- Permutation shapes:** exactly one descent $2^n - n - 1$; bitonic (up then down) / "place smallest at an end" 2^{n-1} ; increasing around a circle (either way) $\frac{2}{(n-1)!}$ prob.
- Parity grids:** $m \times n$ 0/1 with all row/col sums even: $2^{(m-1)(n-1)}$ (last row/col forced). 1..9 in 3×3 , all lines odd $\Leftrightarrow$ odds form a "cross": $\frac{9}{5}$.
- Number theory:** $d(\prod p^e) = \prod (e + 1)$, odd $\Leftrightarrow$ square; # odd divisors odd $\Leftrightarrow$ square or 2-square; random divisor $\Rightarrow$ exponents independent uniform on $0..a$; $v_p(n!) = \sum_i \lfloor n/p^i \rfloor$; #digits = $\lfloor \log_{10} N \rfloor + 1$, $\log_{10} n! \approx n \log_{10} \frac{n}{e} + \frac{1}{2} \log_{10}(2\pi n)$; closest factor pair: largest divisor $\leq \sqrt{N}$. Divisible by 3/9: digit sum; 11: alternating sum; 4/8: last 2/3 digits; 5: last digit.
- Sum over a set** = size $\times$ average (each digit in each position equally often; pair x with complement x').
- Grids:** squares in $n \times n \sum k^2$; rectangles $\binom{n+1}{2}^2$; spanning trees of K_n : n^{n-2} . **Partitions** of set into k blocks $S(n, k)$ (onto = $k!S$); Bell 1, 1, 2, 5, 15, 52, 203; integer partitions 1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42. **Burnside** necklaces (k colours, n beads, rotations): $\frac{1}{n} \sum_{d|n} \varphi(d) k^{n/d}$.
- Number facts:** Josephus (every 2nd): $n = 2^m + L \Rightarrow$ survivor $2L + 1$; coin problem a, b coprime: largest non-representable $ab - a - b$; Pick $A = I + \frac{b}{2} - 1$; $a^{p-1} \equiv 1 \pmod{p}$; $\varphi(n) = n \prod (1 - \frac{1}{p})$; last digits: work mod 10^k , cycles of powers.

3 • Probability rules

- Bayes: posterior $\propto$ prior $\times$ likelihood (equal priors cancel). Total prob over first step / hidden type / which coin.
- Equally likely space:** count faces not coins (one side H $\Rightarrow$ two-headed $\frac{2k-1}{2k+m}$); random cube then random face up $\Rightarrow$ every small face equally likely; revealer's rule matters (Monty $\frac{2}{3}$; n doors, 1 opened: switch $\frac{n-1}{n(n-2)}$; " ≥ 1 boy" $\frac{1}{3}$, "older boy" $\frac{1}{2}$; "boy born Tue" $\frac{2k-1}{4k-1}$ ($k = 7$)).
- Given counts** (h H in n): all $\binom{n}{h}$ arrangements equal $\Rightarrow$ PIE over block start positions. **Given first ace at k :** before = uniform non-aces, after = uniform order.
- "Last one decides":** any sum where one term is uniform mod m : $P(m \mid S) = \frac{1}{m}$; one fair coin among biased: $P(\text{odd heads}) = \frac{1}{2}$; n coins $P(\text{even}) = \frac{1+\prod (q_2 - p_1)}{2}$.
- Rate constant:** $P(\text{none in } \alpha t) = P(\text{none in } t)^\alpha$.
- Simpson:** better in every group, worse overall – possible (different weights).

4 • Expectation & variance

- $\mathbb{E}X = \sum_{k \geq 1} P(X \geq k)$; $\mathbb{E}X = \int_0^\infty P(X > t) dt$; $\mathbb{E}X^2 = \sum (2k - 1) P(X \geq k)$. Law of total exp. $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X|Y)]$.
- Discrete max/min** of n iid on $1..m$: $P(\max \leq k) = (\frac{k}{m})^n$, $P(\min \geq k) = (\frac{m-k+1}{m})^n$, $\mathbb{E} \min = \sum_k (\frac{m-k+1}{m})^n$.
- $\max + \min = X + Y$; $\max = \frac{X+Y+|X-Y|}{2}$. $\mathbb{E}|X - Y|$, iid uniform $1..m$: $\frac{m-1}{3m} (P(|X - Y| = d) = \frac{2(m-d)}{m^2})$.
- $\mathbb{E}[XY] = \mathbb{E}X \mathbb{E}Y$ if indep.; product of indep. factors: $\mathbb{E} \prod = \prod \mathbb{E}$ (stock $\times u$ w.p. p , $\times d$:

- $S_0(pu + qd)^n$. Squared distance of random steps: $\mathbb{E}|\sum v_i|^2 = \sum \mathbb{E}|v_i|^2$ when $\mathbb{E}v_i = 0$. Centre coordinates first (area², determinant $\Rightarrow \mathbb{E} = 0$ by row swap).
- $\text{Var} = \mathbb{E}X^2 - (\mathbb{E}X)^2$; Cov bilinear, $\text{Var} \sum = \sum \text{Var} + 2 \sum_{i < j} \text{Cov}$; counts $\sum I_i$: $\text{Var} = \sum p_i(1 - p_i) + \sum_{i \neq j} (p_{ij} - p_i p_j)$ (fixed points: 1). $\text{Var } X = \mathbb{E} \text{Var}(X|Y) + \text{Var } \mathbb{E}(X|Y)$; random sum $\text{Var } S_N = \mathbb{E} N \text{Var } X + \text{Var } N(\mathbb{E} X)^2$.
- Wald** $\mathbb{E}S_T = \mathbb{E}T \cdot \mathbb{E}X$. **Swap trick:** 2nd-last roll before sum first exceeds N has the law of one roll.

- Waiting for all** of exclusive per-trial events p_i : $\mathbb{E}T = \sum_{j \neq 0} \frac{(-1)^{j+1}}{p_j} \text{ (two: } \frac{1}{p_1} + \frac{1}{p_2} - \frac{1}{p_1 p_2})$.
- Coupons** (n equally likely): all nH_n ; k specified nH_k ; first k distinct $n(H_n - H_{n-k})$; m draws: $\mathbb{E}$ distinct $n(1 - (1 - \frac{1}{n})^m)$, $P(\text{all}) = \frac{\text{onto}(m, n)}{n^m}$. First repeat on m -die: $\mathbb{E} = \sum_{k=0}^m \frac{m!}{(m-k)! m^k} \approx \sqrt{\pi m/2} + \frac{3}{2}$.
- Special cards** (K specials among N , random order): K specials cut others into $K + 1$ gaps of mean $\frac{N-K}{K+1}$; position of j th special $\frac{j(N+1)}{K+1}$; each non-special precedes all specials w.p. $\frac{1}{K+1}$. Multi-stage stop rules: apply gap lemma per stage on the remaining deck.
- Races:** first to k wins vs m : $P(\text{Bin}(k + m - 1, p) \geq k)$; first-head times $A \sim \text{Geo}(p_1)$, $B \sim \text{Geo}(p_2)$: $P(A < B) = \frac{p_1 q_2}{p_1 q_2 + q_1 p_1}$, $\min \sim \text{Geo}(1 - q_1 q_2)$; fair coins $P(A < B < C) = \frac{1}{2^3}$; you strictly first among n fair flippers: $\sum_k 2^{-nk} = \frac{1}{2^n - 1}$.
- Markov $P(X \geq a) \leq \frac{\mathbb{E}X}{a}$; Chebyshev $\leq \frac{\sigma^2}{a^2}$; Jensen convex $\mathbb{E}g(X) \geq g(\mathbb{E}X)$.
- Balls $\rightarrow$ bins** (m into n): bin empty $(1 - \frac{1}{n})^m$;

$\mathbb{E}$ empty $n(1 - \frac{1}{n})^m$; $\mathbb{E}$ colliding pairs $\binom{n}{2}$; birthday $P(\text{some match}) \approx 1 - e^{-m^2/2n}$ (23 of 365 $> \frac{1}{2}$); rare events $P(\text{none}) \approx e^{-\sum p_i}$ (Poisson paradigm).

- Symmetric conditioning:** iid X_i , given $\sum = s$: $\mathbb{E}[X_1 | \cdot] = \frac{s}{n}$. $\mathbb{E}$ #pairs with a property = $\binom{n}{2} p$. Dice: $\mathbb{E} \max$ of n m -dice = $m - \sum_{k=1}^{m-1} (\frac{k}{m})^n$ (2d6: $\frac{161}{36}$, min $\frac{91}{36}$); $\text{Var}(\text{sum of } n \text{ dice}) = \frac{35n}{12}$.
- Sampling w/o replacement** n of N values: $\mathbb{E}$ sum = $n\bar{x}$, $\text{Var} = n\sigma^2 \frac{N-n}{N-1}$. Multinomial $\text{Cov}(N_i, N_j) = -np_i p_j$.

5 • Distributions

	$P(k) / f(x)$	$\mathbb{E}$	Var	MGF
$\text{Bin}(n, p)$	$\binom{n}{k} p^k q^{n-k}$	np	npq	$(q + pe^t)^n$
$\text{Geo}(p)$	$q^{k-1} p$	$\frac{1}{p}$	$\frac{q}{p^2}$	$\frac{pe^t}{1 - qe^t}$
$\text{NegBin}(r, p)$	$\binom{k-1}{r-1} p^r q^{k-r}$	$\frac{r}{p}$	$\frac{rq}{p^2}$	$(\cdot)^r$
$\text{Po}(\lambda)$	$e^{-\lambda} \frac{\lambda^k}{k!}$	λ	λ	$e^{\lambda(e^t - 1)}$
$\{1..n\}$	$\frac{1}{n}$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$	
$\mathbb{U}[a, b]$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt}-e^{at}}{t(b-a)}$
$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{e^{-\lambda t}}{1 - \lambda t}$
arrival T_k	$\frac{\lambda^k e^{-\lambda} k!}{(k-1)!}$	$\frac{k}{\lambda}$	$\frac{k}{\lambda^2}$	$(\frac{e^{-\lambda t}}{1 - \lambda t})^k$
$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$e^{i\mu t - \frac{1}{2} \sigma^2 t^2}$

- Links:** $\text{Bin}(n, p) \approx \text{Po}(np)$ (n big, p small); $\approx N(np, npq)$ ($\pm 1\%$); sum of indep. Po / Bin (same p) / normal stays in family (parameters add); $T_k =$ sum of k iid Exp , $P(T_k \leq t) = P(\text{Po}(\lambda t) \geq k)$; $[\text{Exp}]$ is geometric; $-\frac{1}{\lambda} \ln U \sim \text{Exp}(\lambda)$; $F^{-1}(U) \sim F$, $F(X) \sim U$.
- Bin:** mode $\lfloor (n+1)p \rfloor$; $P(\text{even}) = \frac{1+(q-p)^n}{2}$; $\mathbb{E}X^2 = npq + n^2 p^2$; $\mathbb{E}[X(n - X)] = n(n - 1)pq$. **Po:** $\mathbb{E}[X(X - 1)] = \lambda^2$; $P(\text{even}) = \frac{1+e^{-2\lambda}}{2}$; given $X + Y = n$: $X \sim \text{Bin}(n, \frac{\lambda_1}{\lambda_1 + \lambda_2})$.
- Tails:** $P(\text{Geo} > k) = q^k$ (trials), $P(\text{Exp} > t) = e^{-\lambda t}$; failures-before-success version has mean $\frac{p}{1 - qP}$. **Memoryless:** only Geo, Exp : $P(X > s + t|X > s) = P(X > t)$; $\mathbb{E}[X|X > a] = a + \mathbb{E}X$.
- Density toolbox:** $f = F'$; $P(a \leq X \leq b) = \int_a^b f$; $\mathbb{E}g(X) = \int gf$; $Y = g(X)$: find $P(Y \leq y)$ then differentiate (monotone: $f_Y = f_X |\frac{dx}{dy}|$); max of n iid: F^n , density $nF^{n-1}f$; min: $1 - (1 - F)^n$, density $n(1 - F)^{n-1}f$; k th smallest density $\frac{n!}{(k-1)!(n-k)!} F^{k-1}(1 - F)^{n-k} f$.

- Joint:** indep. $\Leftrightarrow f(x, y) = f_X f_Y$; marginal $f_X = \int f dy$; conditional $f(x|y) = \frac{f(x, y)}{f_Y(y)}$; $P(A) = \int P(A|x) f(x) dx$; $X|X < Y$: density $\propto f(x) P(Y > x)$. Sum: $f_{X+Y}(z) = \int f_X(x) f_Y(z - x) dx$ (U+U triangle, Exp+Exp $\lambda^2 z e^{-\lambda z}$); ratio U/U : $\frac{1}{2}$ on $[0, 1]$, $\frac{1}{2z^2}$ beyond. Polar: $dx dy = r dr d\theta$.
- Handy conditionals:** $\mathbb{E}[\text{Bin} \mid \geq 1] = \frac{np}{1 - q^n}$; $\mathbb{E}[\text{Po} \mid \geq 1] = \frac{\lambda}{1 - e^{-\lambda}}$; $\mathbb{E}[\frac{1}{1 + \text{Po}(\lambda)}] = \frac{1 - e^{-\lambda}}{\lambda}$; $\frac{1}{\mathbb{E}[1 + \text{Bin}(n, p)]} = \frac{1 - q^{n+1}}{(n+1)p}$, $P(\text{Geo even}) = \frac{q}{1+q}$; $\mathbb{E} \max$ of 2 iid $\text{Geo}(p) = \frac{2}{p} - \frac{1}{1 - q^p}$. Order stat $\leftrightarrow$ binomial: $P(U_{(k)} \leq x) = P(\text{Bin}(n, x) \geq k)$.

6 • Uniform

- $\mathbb{E}U^k = \frac{1}{k+1}$, $\text{Var } U = \frac{1}{12}$. n iid: $\mathbb{E}U_{(k)} = \frac{k}{n+1}$ (n points cut $[0, 1]$ into $n + 1$ exchangeable gaps), $\max \frac{n}{n+1}, \min \frac{1}{n+1}$, range $\frac{n+1}{n+1}$; $\text{Var } U_{(k)} = \frac{k(n+1-k)}{(n+1)^2(n+2)}$; given $\max = m$, rest iid $\mathbb{U}[0, m]$.
- Two:** $\mathbb{E} \max = \frac{2}{3}$, $\mathbb{E} \min = \frac{1}{3}$, $\mathbb{E}|U - V| = \frac{1}{3}$ (on $[0, 1]$: $\frac{2}{3}$), $\mathbb{E}(U - V)^2 = \frac{1}{6}$; $P(|U - V| \leq t) = 1 - (1 - t)^2$; $P(U + V \leq t) = \frac{t^2}{2} (t \leq 1)$; $P(UV \leq c) = c - c \ln c$; $P(1 \leq \frac{U}{V} \leq 2) = \frac{1}{4}$; $P(\max \leq 2 \min) = \frac{1}{2}$; $\max(\mathbb{U}, 1 - U) \sim \mathbb{U}[\frac{1}{2}, 1]$; $\mathbb{E} \max(U, c) = \frac{1+c^2}{2}$; $\mathbb{E}|U|U > a| = \frac{1+a}{2}$.
- Two points in** (a, b) straddle c : $2 \frac{(c-a)(b-c)}{(b-a)^2}$. Coin diameter d inside square cell s : $(\frac{d}{s})^2$. $a_k \sim \mathbb{U}[0, k]$ ordered: volume of ordered region / box volume.
- Stick into n pieces** ($n - 1$ cuts): exchangeable, each mean $\frac{1}{n}$; $P(\text{given } k \text{ pieces all } > x) = (1 - kx)^{n-1}$; $\mathbb{E}$ longest $\frac{H_n}{3}$ (3 pieces $\frac{11}{18}$), shortest $\frac{1}{n^2}$; polygon possible $1 - \frac{1}{2^{n-1}}$ (triangle $\frac{1}{4}$); one cut: shorter $\sim \mathbb{U}[0, \frac{1}{2}]$, longer $\sim \mathbb{U}[\frac{1}{2}, 1]$.
- Circle, n points:** all in some semicircle $\frac{n-1}{n}$ (3 pts: obtuse $\frac{3}{4}$, acute/contains centre $\frac{1}{4}$); $\mathbb{E}$ #points until not in a semicircle $= 1 + \sum_{n \geq 1} \frac{1}{2^{n+1}} = 5$; 4 points: 3 pairings, 1 crosses $\Rightarrow$ two random chords cross $\frac{1}{3}$; arcs all $\geq a$: $(1 - na)^{n-1}$. Discrete N points on circle, obtuse: count by first vertex clockwise.
- Sums:** $P(U_1 + \dots + U_n \leq t) = \frac{t^n}{n!} (t \leq 1) \Rightarrow$ # U to exceed 1: $P(N > n) = \frac{1}{n!}$, $\mathbb{E} N = e$, $\mathbb{E}$ final sum $\frac{e}{2}$ (Wald). **Repeated uniform cuts:** keep U fraction until < 1 from L : $\mathbb{E}$ #cuts = $1 + \ln L$ ($-\ln U \sim \text{Exp}(1)$).
- Only order matters** $\Rightarrow$ discrete: 4 iid points, segments $P_1 P_2, P_3 P_4$ overlap $\frac{2}{3}$; k of n points owned by one person: positions uniform over $\binom{n}{k}$.
- Shapes:** uniform in disc: $P(R \leq r) = r^2$, $\mathbb{E}R = \frac{2}{3}$; sphere surface: each coord $\mathbb{U}[-1, 1]$, $\text{Var} = \frac{1}{3}$; in triangle height h : $P(D > t) = (1 - \frac{h}{3})^2$, $\mathbb{E}D = \frac{4}{3}$.
- Geometry constants:** two random points on circle (circumference L) short-arc distance $\mathbb{E} = \frac{L}{\pi}$; random chord (2 random points, radius r) $\mathbb{E} = \frac{4r}{\pi}$; Buffon needle $\ell \leq d$: $\frac{2\ell}{\pi}$; $ax^2 + bx + c$ with $a, b, c \sim \mathbb{U}$ real roots $\frac{3\pi \ln 4}{36}$; random triangle in triangle $\mathbb{E}$ area $\frac{1}{12}$.

7 • Poisson process & exponential

- Rate λ : $N(t) \sim \text{Po}(\lambda t)$, disjoint intervals independent, gaps iid $\text{Exp}(\lambda)$. $P(\text{none in } t) = e^{-\lambda t}$; first arrival density $\lambda e^{-\lambda t}$; time of k th: table.
- Merge:** rates add; next event is type i w.p. $\frac{\lambda_i}{\sum \lambda_j}$ (independent of time). **Thin** (each kept w.p. p): kept $\text{Po}(p\lambda t)$ and dropped $\text{Po}(q\lambda t)$ independent (knowing buyers says nothing about browsers).
- Poissonize:** $\text{Po}(\lambda)$ items into m equal bins $\Rightarrow$ independent $\text{Po}(\frac{\lambda}{m})$: all bins hit ($1 - e^{-\lambda/m}$) ^{m} . Fixed n balls into n bins all hit: $\frac{n!}{e^n}$.
- Given** $N(T) = n$: arrival times = sorted iid $\mathbb{U}[0, T]$: k th at $\frac{kT}{n+1}$ on average.
- Toggle/parity:** light toggled at rate λ for t : $P(\text{even}) = \frac{1+e^{-2\lambda t}}{2}$.

- **Exp races:** $\min_i \text{Exp}(\lambda_i) \leq \text{Exp}(\sum \lambda_i)$, independent of which is min; $P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$; $\mathbb{E} \max$ of n iid $= \frac{H_n}{n}$ (spacings $\text{Exp}(k\lambda)$); quantile: $P(T < t) = \alpha \Rightarrow t = -\frac{\ln(1-\alpha)}{\lambda}$; median $\frac{\ln 2}{\lambda}$.
- **Inspection paradox:** interval containing a random time has mean $\frac{\mathbb{E}G^2}{\mathbb{E}G}$ (Poisson: $\frac{2}{\lambda}$, both sides Exp); wait $= \frac{\mathbb{E}G^2}{2\mathbb{E}G}$. Fixed gaps g : wait $U[0, g]$, mean $\frac{g}{2}$. Gaps a w.p. $1 - \epsilon$, b w.p. ϵ : $\frac{(1-\epsilon)a^2 + \epsilon b^2}{2(1-\epsilon)a + \epsilon b}$.
- **Window each cycle** (board only during fraction w of period T): $\mathbb{E} \text{wait} = \frac{(1-w)^2 T}{2}$.
- **Renewal:** running die total hits n w.p. $\rightarrow \frac{1}{n} = \frac{2}{n}$; overshoot of first total $\geq N$: $P(N+j) \approx \frac{m-1}{m\mu}$ ($j = 0..m-1$); stop at first odd sum with fair m -die (m even): $P(S)$ grows by $\frac{m-1}{m}$ per odd step up to $m-1$, then falls $\Rightarrow$ mode $m-1$.

8 - Normal

- Standardise $Z = \frac{X-\mu}{\sigma}$; $P(a \leq X \leq b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$; $\Phi(-z) = 1 - \Phi(z)$. Indep. $aX + bY \sim \mathcal{N}(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$ (difference: variances add); $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$; CLT $S_n \approx \mathcal{N}(n\mu, n\sigma^2)$, integer $\Rightarrow \pm \frac{1}{2}$.
- **Moments** ($X \sim \mathcal{N}(0, \sigma^2)$): odd $= 0$; $\mathbb{E}X^2 = \sigma^2$, $\mathbb{E}X^4 = 3\sigma^4$, $\mathbb{E}X^{2m} = (2m-1)!!\sigma^{2m}$; $\text{Var } X^2 = 2\sigma^4$; $\mathbb{E}|X| = \sigma\sqrt{2/\pi}$; $\mathbb{E}X^2$ general $= \sigma^2 + \mu^2$.
- **Exponential moments:** $\mathbb{E}e^{tX} = e^{\mu t + \sigma^2 t^2/2}$ (lognormal mean $e^{\mu t + \sigma^2 t^2/2}$, $\text{Var} = (e^{\sigma^2} - 1)e^{2\mu t + \sigma^2}$); **tilt** $\mathbb{E}[e^{tX} \mathbf{1}_{X>a}] = e^{\mu t + \sigma^2 t^2/2} \Phi(\frac{\mu - \sigma^2 t}{\sigma})$.
- **Truncation:** $\mathbb{E}[Z|Z > a] = \frac{\varphi(a)}{1-\Phi(a)}$; $\mathbb{E}(Z - a)_+ = \varphi(a) - a(1 - \Phi(a))$; $\mathbb{E} \max(Z, c) = c\Phi(c) + \varphi(c)$.
- **Two iid** $\mathcal{N}(\mu, \sigma^2)$: $\mathbb{E}|X - Y| = \frac{2\sigma}{\sqrt{\pi}}$, $\mathbb{E} \max = \mu + \frac{\sigma}{\sqrt{\pi}}$, 3 iid: $\mathbb{E} \max = \mu + \frac{3\sigma}{2\sqrt{\pi}}$.

Joint density indep.: $\frac{1}{2\pi\sigma_1\sigma_2} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2} - \frac{(y-\mu_2)^2}{2\sigma_2^2}}$.

- **Conditioning on a sum:** $\mathbb{E}[X_1|X_1 + X_2 = s] = \mu_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}(s - \mu_1 - \mu_2)$. Bivariate corr. ρ : $Y = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(X - \mu_X) + \sigma_Y \sqrt{1 - \rho^2} Z'$ $\Rightarrow \mathbb{E}[Y|X]$ linear, $\text{Var}(Y|X) = \sigma_Y^2(1 - \rho^2)$.
- **Rotation invariance:** iid $\mathcal{N}(0, 1)$ pair has uniform angle $\Rightarrow$ wedge probability $= \frac{\text{angle}}{2\pi}$; $P(X, Y > 0) = \frac{1}{4} + \frac{\arcsin \rho}{2\pi}$ ($\rho = \frac{1}{2}$: $\frac{1}{3}$); $R^2 = X^2 + Y^2 \sim \text{Exp}(\frac{1}{2})$; $P(X > Y|X > 0) = \frac{3}{4}$. Stein: $\mathbb{E}[Zg(Z)] = \mathbb{E}g'(Z)$.

9 - Coins, dice, patterns

Conway (any pattern): $\mathbb{E}T = \sum_L \text{prefix}_L = \text{suffix}_L \cdot \frac{1}{P(\text{prefix}_L)}$; fair coin $\sum 2^L$: HH 6, HT 4, HHH 14, HTH 10, THH 8, HHTT 20. **Penney** $AB = \sum 2^L$ over $\text{suf}_L(A) = \text{pre}_L(B)$: $P(A \text{ before } B) = \frac{BB-BA}{(AA-AB)+(BB-BA)}$. Any HHH...H not at the start is preceded by T $\Rightarrow$ THH beats HHH w.p. $\frac{7}{8}$.

- k successes in a row: $\mathbb{E} = \frac{2^{k+1}-1}{2}$ (fair $2^{k+1} - 2$); k equal in a row m -die: $\frac{m^{k+1}-1}{m-1}$; a specific face twice in a row $\frac{1+m}{p^2}$.
- First HH at toss n : $\frac{F_{n+1}}{2}$; no HH in n : $\frac{F_{n+2}}{2}$; ends on even toss $\frac{3}{5}$. Toss game "HT, who tosses T wins": states prev H/not: $x = \frac{3}{5}$, starter $\frac{4}{9}$.
- **Exactly a run of k in n :** fix start, neighbours T, rest free, PIE for 2 runs. $\mathbb{E}\# \text{runs} = 1 + 2pq(n-1)$; $\mathbb{E}\# \text{HT} = \frac{n-1}{4}$; $\# \text{HT} - \# \text{TH} = [\text{first H}] - [\text{last H}]$.
- $n+1$ vs n fair coins, more heads: $\frac{1}{2}$. Heads $\times$ tails in n : $\mathbb{E} = \frac{n(n-1)}{4}$.
- **Dice sums:** 2 dice $P(S = s) = \frac{6-s-7}{36}$, $P(\max = k) = \frac{2k-1}{36}$; $\#$ ways k dice sum s : $\sum_j (-1)^j \binom{k}{j} \binom{s-6j-1}{k-1}$; sorted n rolls $\binom{n+5}{5}/6^n$. Top-3 of n dice $= 18 \Leftrightarrow \geq 3$ sixes: $P(\text{Bin}(n, \frac{1}{6}) \geq 3)$.
- **Roll until 6:** $\mathbb{E}$ rolls 6, sum 21 (Wald); sum even $\frac{1}{2}$ ($p = \frac{1}{6} + \frac{1}{2}p + \frac{1}{2}(1-p)$); given no 5 before the 6: time to first of $\{5, 6\}$ is $\text{Geo}(\frac{2}{3})$ whichever wins $\Rightarrow 3$. Counts of faces before first 6: ignore irrelevant faces $\Rightarrow$ smaller fair die.
- **Roll until different** (m -die): $\mathbb{E} \text{sum} = \mu \frac{2m-1}{m-1}$; roll until repeat: first-repeat formula $\S 4$.

Consecutive rolls within 1 (die): 3 symmetric states $i \leftrightarrow 7 - i$.

- **Painted n^3 cube:** corners 8 (3 faces), edges $12(n-2)(2)$, faces $6(n-2)^2(1)$, inner $(n-2)^3$; red small faces $6n^2$ of $6n^3 \Rightarrow$ random cube rolled shows red $\frac{1}{n}$; $\mathbb{E}\#$ red faces $\frac{6}{n}$.
- **Series / tennis:** best-of- $(2n-1)$: $\sum_{k=0}^{n-1} \binom{n-1+k}{k} p^n q^k$; game from 0-0: $p^4(1+4q + \frac{1}{1-q^2}) + 20p^3q^3 \frac{p^2}{p^2+q^2}$, from deuce/30-30 $\frac{p^2}{p^2+q^2}$. Craps win $\frac{244}{495}$. De Méré: ≥ 1 six in 4 vs double six in 24.
- $\mathbb{E}$ flips for n heads total $\frac{n}{p}$; $\mathbb{E}\# \text{HH}$ (overlapping) in n $\frac{n+1}{4}$; longest run in $n \approx \log_2 n$; k th success exactly at trial t : NegBin.

10 - Random walks & gambler's ruin

Start s , stop at 0 or N ; up p , down q , $r = \frac{a}{p}$. $P(\text{hit } N)$: fair $\frac{s}{N}$; biased $\frac{1-r^N}{1-r}$. $\mathbb{E}T$: fair $s(N - s)$; biased $\frac{N P(N) - s}{p - q}$. **Conditioned on hitting N** (fair): $\mathbb{E}T = \frac{N^2 - s^2}{3}$; on 0: $\frac{s(2N-s)}{3}$. Step sizes ± 1 with stay prob c : $\mathbb{E}T \div (1 - c)$. **One barrier** ($+k$ above): $P = \min(1, (p/q)^k)$, $\mathbb{E}T = \frac{k}{p-q}(p > q)$; fair: $P = 1$, $\mathbb{E}T = \infty$. Targets $-a, +b$: replace $s \rightarrow a$, $N \rightarrow a + b$.

- **Martingales** (fair game stays fair when stopped): S_n (fair), $S_n - n\mu$, $S_n^2 - n\sigma^2$, r^{S_n} , $e^{\theta S_n}$ with $\mathbb{E}e^{\theta X} = 1$, fraction of red cards left. OST valid if stopped process bounded or $\mathbb{E}T < \infty$ with bounded steps. 2-D walk leaving disc R : $\mathbb{E}T = \mathbb{E}|S_T|^2 \approx R^2$.
- **Casino trick** gives Conway; **remaining-fraction** martingale $\Rightarrow$ any stopping rule for "next card red" wins $\frac{1}{2}$; "next two same colour" = "last two" $\frac{n-1}{2n-1}$.
- $P(S_{2n} = 0) = \binom{2n}{n} 4^{-n} = P(\text{no return to 0 in } 2n)$; first return at $2n$: $\frac{C_{n-1}}{2^{2n-1}}$; $P(\max_{k \leq n} S_k \geq m) = 2P(S_n > m) + P(S_n = m)$.
- **Walk on graph** (uniform neighbour): $\pi_v = \frac{\deg v}{2|E|}$, return to v : $\frac{2|E|}{\deg v}$; path 0.. N end to end N^2 ; from k to end $N^2 - k^2$; cycle n , distance k : $k(n-k)$; K_n to a given vertex $n-1$; cube to opposite corner 10; tree child $\rightarrow$ parent 2 $\cdot |\text{subtree}| - 1$; commute $= 2|E| R_{\text{eff}}$. Lump states by distance (icosahedron, tetrahedron, cube).
- Back at start after n steps on K_m : $\frac{1}{m} + (1 - \frac{1}{m})(\frac{m-1}{m})^n$.

11 - Markov chains

- n -step: P^n ; path prob $= \text{start} \times \text{product of transitions}$. Absorption $a_i = \sum_j P_{ij} a_j$ ($a = 1$ at target, 0 at other absorbers); time $t_i = 1 + \sum_j P_{ij} t_j$.
- Stationary $\pi P = \pi$, $\sum \pi = 1$; detailed balance $\pi_i P_{ij} = \pi_j P_{ji}$; doubly stochastic $\Rightarrow$ uniform. **Kac:** $\mathbb{E}$ return time to $i = \frac{1}{\pi_i}$ (k bulbs, random one toggled: 2^k).
- Two-state (leave 1 w.p. a , leave 2 w.p. b): $\pi = (\frac{b}{a+b}, \frac{a}{a+b})$, $p_{11} = \frac{b+a(1-a-b)}{a+b}$, hit 1 from 2: $\frac{1}{b}$.
- If the next-step law is the same from every state $\Rightarrow$ iid $\Rightarrow$ geometric wait. Transient/recurrent: closed class = recurrent (finite chain).
- **Absorbing chain matrix:** $N = (I - Q)^{-1}$ (N_{ij} = visits to j); times $N\mathbf{1}$; absorption NR .

12 - Random permutations & structures

- Fixed points: mean 1, var 1, $P(k) \rightarrow \frac{e^{-1}}{k!}$, $\#k$ -cycles $\frac{1}{k}$ (2-cycles $\frac{1}{2}$); $\#$ cycles H_n ; given element in cycle of length k : $\frac{1}{k}$; records H_n ; inversions $\frac{n(n-1)}{4}$; local maxima $\frac{n+1}{3}$ (interior $\frac{1}{2}$, end $\frac{1}{2}$); longest zigzag $\frac{2n+1}{3}$ (interior turns $\frac{2}{3}$).
- **Suffix minima** (car clusters: car leads iff slower than all ahead) H_n . Random matching of n left- n right points: crossings $\frac{1}{2} \binom{n}{2}$.
- **Random-order removal game** (pick uniform remaining item, remove it and everything below/its multiples): $\mathbb{E}\# \text{picks} = \sum_v \frac{1}{\#(v \text{ and its ancestors})}$ (tree: $\sum \frac{1}{\text{depth}}$; numbers 1.. n , remove multiples: $\sum_k \frac{1}{\lfloor n/k \rfloor}$).
- **Front person reinserted uniformly at the back $n-1$ slots;** last person reaches front: sum of geometric waits $\sum_{k=1}^{n-1} \frac{1}{k} = nH_{n-1}$.
- **Pólya urn** (add same colour; free throws with make prob = made/taken): from (1, 1) after n

- draws the count is uniform on 0.. n ; draw = current fraction (martingale). Laplace succession $\frac{k+1}{n+2}$.
- **Relabel symmetry:** any rule treating colours alike $\Rightarrow$ each of c colours last w.p. $\frac{1}{c}$ (monochromatic removal). Boarding pass: last seat $\frac{1}{2}$.
- **Random pairing** of $2n$ ends: $\mathbb{E}\# \text{loops} \sum_{k=1}^n \frac{1}{2k-1}$; two given noodles same loop $\frac{2}{n}$. Random graph $G(n, p)$: $\mathbb{E}\# K_k = \binom{n}{k} p^{\binom{k}{2}}$. v vertical, h horizontal lines: $(v+1)(h+1)$ cells; $v \sim \text{Bin}(n, \frac{1}{2})$, $h = n - v$: $\mathbb{E} = (\frac{n}{2} + 1)^2 - \frac{n}{2}$. Marbles: pair drawn together on some move $\frac{\# \text{moves}}{\binom{2}{2}}$.

13 - Stopping, betting, strategy

- **Fair price** = $\mathbb{E}$ payout. **Finite horizon:** $V_1 = \mathbb{E}X$, $V_k = \mathbb{E} \max(X, V_{k-1})$: die 3.5, 4.25, $\frac{14}{3}$, $\frac{55}{18}$; m -die keep $x > V$: $V' = \frac{|V|V + \sum_{i=1}^m i}{m}$; uniform $v' = \frac{1+m}{2}$.
- **Infinite rerolls, cost c :** $V = -c + \mathbb{E} \max(X, V)$, guess threshold, check (die, $c = 1$: keep 4-6, $V = 4$); uniform: $V = 1 - \sqrt{2c}$.
- **Accumulate or bust:** each roll busts w.p. q , else adds mean μ : stop at $s \geq \frac{(1-q)\mu}{q}$; value $\approx s^*(1-q)^{s^*/\mu}$. Extra life $\Rightarrow$ never stop while you hold it.
- **Reroll-with-penalty:** value of reroll after $r = \frac{1}{m} \sum_s \text{payoff}(r, s)$; keep r iff $r \geq$ that.
- **Hunt then harvest** (M moves, m -die, roll until face $\geq n$ then take every move): $\frac{n-2}{m} (M - \frac{m}{m-n+1})$, maximise over n .
- **Slot filling** (digits into places): backwards; put in bigger slot iff above mean of remaining.
- **Compare two actions** as functions of x , solve the inequality (pay c to double vs keep: $\frac{\min(2x-1, m)}{m} - c > \frac{x-1}{m}$).
- **Secretary:** skip $\frac{n}{e}$, then first best-so-far: $P \rightarrow \frac{1}{e}$; n small: maximise $\frac{e}{n} \sum_{k=r}^{n-1} \frac{1}{k}$.
- **Guess colours** (nR , nB , bet majority): $\mathbb{E} \text{correct} = n + \frac{1}{2} (\frac{4^n}{\binom{2n}{n}} - 1)$. Choose-a-suit races: wait one card, take its suit.
- **Information:** pick the most likely hypothesis per observation; impossible values reveal ($X/2X$: odd $\Rightarrow$ small, above max $\Rightarrow$ big; win $\frac{3}{4}$ for $X \sim \{1..N\}$); $X/2X/3X$ wanting $3X$: not a multiple of 3 $\Rightarrow$ switch; consecutive envelopes $x + 1..3$: keep 2nd iff it rose, $\mathbb{E} = x + \frac{2}{3}$. Unknown majority (9:1 or 1:9): eat one, guess its flavour for the rest.
- **2x2 zero-sum** $\frac{a}{a-b-c+d}, \frac{b}{d}$, no saddle: row plays top w.p. $\frac{d-c}{a-b-c+d}$, value $\frac{ad-bc}{a-b-c+d}$. Drop dominated strategies first (RPS without rock: $\frac{1}{3}$).
- **Adversary picks your worst** $\Rightarrow$ equalise: payouts i w.p. p_i : $p_i \propto \frac{1}{i}$, loss $\frac{1}{\sum \frac{1}{i}}$; integer deck (D cards, pays i if named i drawn): $k_i \leq \lfloor M/i \rfloor$, smallest M with $\sum_i \lfloor M/i \rfloor \geq D$, loss $\frac{M}{D}$.

- **Location games:** best response sits adjacent on the richer side $\Rightarrow$ leader balances sides ($\sum x_{\leq a} x \approx \sum x_{> a} x$, i.e. $a \approx \sum$). Circle, 3 players: 2nd opposite 1st, 3rd uniform. Pipe $\frac{3}{8}$. Larger-pays-smaller: optimum $b \approx \frac{N}{3}$, $\mathbb{E} = \frac{b(2N+1-3b)}{2N}$.
- **Keep-or-redraw duel** (uniform): cutoff t solves $F(t) = \text{value of redraw} \Rightarrow t^2 + t - 1 = 0$, $t = \frac{\sqrt{5}-1}{2}$.
- **Pass games:** mover's value r ; passing gives $-r \Rightarrow$ accept $k \geq -r$, solve r self-consistently.
- **Kelly** (odds $b:1$): $f^* = p - \frac{q}{b}$. Bold play in fair game: $P(\text{reach goal}) = \frac{\text{wealth}}{\text{goal}}$.

14 - Two-player games

- P-position (loser to move): every move $\rightarrow$ N; N: some move $\rightarrow$ P. Tabulate small n , spot period. Take 1.. k : lose $\Leftrightarrow (k+1) \mid n$; take $\{1, 3, 4\}$: lose $\Leftrightarrow n \bmod 7 \in \{0, 2\}$; take $\leq$ half: lose $\Leftrightarrow n = 2^j - 1$; subtract a proper divisor: odd loses; Nim: XOR = 0 loses; king to corner: (even, even) loses.
- **Mirror** (centre, diameter, equal halves of a line, reflected row). **Pairing:** mn odd board, piece to unvisited neighbour $\Rightarrow$ 2nd player wins. **Strategy stealing / waiting move.** **Fixed number of moves** (each move +1 pile: moves

- $= \text{total} - \# \text{piles}$ $\Rightarrow$ parity decides. Coin row: $b(l, r) = \text{sum}(l..r) - \min(b(l+1, r), b(l, r+1))$.
- **Sprague-Grundy:** $g(s) = \text{mex}\{g(s')\}$; sum of games: XOR of g ; lose $\Leftrightarrow 0$. Misère Nim: as Nim unless all heaps ≤ 1 (then odd $\#$ heaps loses). Symmetric strategy stealing $\Rightarrow$ first player can't lose (Chomp, Hex).

15 - Puzzle toolkit

- **Double counting:** sum of all pair sums $= (n-1) \times \text{total}$; splitting n into piles scoring xy always totals $\binom{n}{2}$.
- **Invariants:** parity of sum; alternating sum; $\prod (x_i + 1)$ for $a, b \rightarrow ab + a + b$ (final $\prod (x_i + 1) - 1$); $|a - b|$: sum parity fixed; black-white sum under neighbour $\pm$; $\sum \text{pos}^2$ monovariant; perimeter never grows (infection needs $\geq n$ cells).
- **Pigeonhole:** $2^n - 1$ nonempty subsets vs $\leq n$ $\cdot$ max sums $\Rightarrow$ two disjoint equal-sum subsets; 5 lattice points $\Rightarrow$ integer midpoint (4 parity classes); friends counts 0 and $n-1$ can't coexist.
- **Colouring:** chessboard (domino covers 1B1W; each step flips colour); $(i + j) \bmod k$ for $1 \times k$ tiles.
- **Mod-2 linear algebra:** toggles commute and cancel $\Rightarrow k$ moves give $\leq 2^k$ states ($< 2^{\text{cells}} \Rightarrow$ not always solvable); row/col add: need target $= r_i + c_j$.
- **Divisibility/integers:** fractions of a crowd $\Rightarrow$ lcm divides; iterate $x \mapsto ax + b$ n times: $a^n x + b \frac{a^n-1}{a-1}$, then minimise over integers; $|\frac{a}{b} - \frac{1}{2}| \leq \epsilon$: even b always, odd b needs $\frac{1}{b} \leq \epsilon$. Pile doubling $(x, N - x)$: $x \mapsto 2x \bmod N$, empty $\Leftrightarrow N = 2^j$. p^2 square $\Rightarrow p$ even or square. Self-referential "at most $n-1$ true": $T = 100 - T$.
- **Hats:** parity bets (half say even, half odd $\Rightarrow$ one half all right); 3 players: speak iff the two seen match $\Rightarrow \frac{3}{4}$; 2-1 players (Hamming): $\frac{n}{n+1}$.
- **Weighing:** take k coins from box k : deficit names box; balance: $3^w \geq \# \text{outcomes}$. **Eggs:** e eggs, d drops cover $f(d, e) = \sum_{k=1}^e \binom{d}{k}$ floors (2 eggs: $\frac{d(d+1)}{2} \geq \text{floors}$). **Horses** 25 by fives, top 3: 7 races (5 heats, winners, one race of candidates). **Transport** (eat 1/mile per load, capacity C): cost/mile $= \lceil \text{stock}/C \rceil$, drop to next level after spending C .
- **Information content:** $\# \text{messages} = \prod (\text{choices per part}) \Rightarrow \lfloor \log_2 \rfloor$ bits (piece broken on day 1.. D or never: $(D+1)^n$).
- **Queues:** your expected spot among m randoml ordered $= \frac{m+1}{2}$.
- **Extremal / induction:** closest pair shoot each other (odd $n \Rightarrow$ someone dry); gas cans: some can reaches the next, merge; add a line, flip one side.

16 - Sums, integrals, approximations

- $\sum_{k=0}^{n-1} x^k = \frac{1-x^n}{1-x}$; $\sum_{k \geq 1} kx^{k-1} = \frac{1}{(1-x)^2}$; $\sum k^2 x^{k-1} = \frac{1+x}{(1-x)^3}$; $\sum_{k \geq 0} \binom{r}{k} x^k = \frac{1}{(1-x)^{r+1}}$; $\sum_{k \geq 1} \frac{x^k}{k} = -\ln(1-x)$; $\sum \frac{x^k}{k!} = e^x$.
- $\sum k = \frac{n(n+1)}{2}$, $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum k^3 = (\sum k)^2$, $\sum \frac{n(n+1)}{k(k+1)} = \frac{n+1}{n}$; $\sum \frac{1}{k^2} = \frac{\pi^2}{6}$; $\sum \frac{(-1)^{k+1}}{k} = \ln 2$.
- $\int_0^1 x^a (1-x)^b = \frac{a!b!}{(a+b+1)!}$; $\int_0^\infty x^n e^{-\lambda x} = \frac{n!}{\lambda^{n+1}}$; $\int e^{-ax^2+bx} = \sqrt{\frac{\pi}{a}} e^{b^2/4a}$; $\int \ln x = x \ln x - x$; $\int_0^1 x^a \ln x = -\frac{1}{(a+1)^2}$. Simplex $\{x_i \geq 0, \sum \leq t\}$: $\frac{t^n}{n!}$. Region probability: draw it, integrate inner variable over its allowed range.
- $H_n \approx \ln n + 0.5772$; $n! \approx \sqrt{2\pi n} (\frac{n}{e})^n$; $\binom{2n}{n} \approx \frac{4^n}{\sqrt{\pi n}}$; $(1 - \frac{x}{n})^n \approx e^{-x}$; $P(\text{no repeat in } k \text{ of } m) \approx e^{-k^2/2m}$.

Traps: " k heads in a row" $2^{k+1} - 2$ vs " k equal" $2^k - 1$. "until" includes the last trial, "before" doesn't. strict vs ties (discrete max/min). $\text{Var}(X - Y) = \text{Var } X + \text{Var } Y$: fixed-total split is dependent, Poisson thinning is not. conditioning on which end is hit changes durations. paths are not equally likely unless each step is .respin vs no respin ($\frac{1}{1-q}$ -type vs uniform position). $\mathbb{E} \frac{1}{X} \neq \frac{1}{\mathbb{E}X}$.